

An ordinal Chomp transition at 2

A computer-assisted counterexample to winner stability
and to the minimal-transition question

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Abstract

We refute the winner-stability conjecture for ordinal extensions of numerical-semigroup Chomp, and we answer negatively the sharper question asking whether a first-player win must already occur at the first possible exponent. For each of $S = \langle 4, 5, 6 \rangle$ and $S' = \langle 4, 6, 9 \rangle$, Chomp on the semigroup is a second-player win, whereas Chomp on its ordinal extension below ω^2 is a first-player win. In both cases the least winning opening, in the usual ordering of ordinals, is $\omega \cdot 4 + 4$, and the same move wins at every higher exponent. Consequently $\text{ch}(\{4, 5, 6\}) = \text{ch}(\{4, 6, 9\}) = 2$.

The proof isolates a finite spectral statement: every nonempty-parameter Apéry remainder, with its bottom playable, has Grundy value at least 3. Equivalently, in the poisoned convention, every such remainder with its bottom deleted has Grundy value at least 2. We prove this statement by exact finite-state certificates: nine gap shapes with a seven-row memory for $\langle 4, 5, 6 \rangle$, where the complete state at step 43 repeats at step 51, and seventeen gap ideals with an eleven-row memory for $\langle 4, 6, 9 \rangle$, where the complete state at step 237 repeats at step 265. Induction then covers every later parameter. All 387 and 4301 certificate entries are independently checked by direct, untruncated finite-poset calculations; the 4301 entries for $\langle 4, 6, 9 \rangle$ are checked under both conventions, the 387 for $\langle 4, 5, 6 \rangle$ under the unpoisoned one alone. We give the order-theoretic reduction, two sufficient criteria for the mechanism, a general finite-window theorem, two independent descriptions of the winning responses, both complete certificates, and reproducible Python verifiers with a regression suite.

Keywords. Poset games; numerical semigroups; ordinal powers; Apéry sets; Sprague–Grundy theory; finite-state certificates.

Status and scope. This is a computer-assisted research result, not a peer-reviewed publication or a proof-assistant formalization. It has not been externally refereed. The finite computations reported here were executed and cross-checked by separate implementations; “independently” means code-level independence, not an external human replication. The infinite conclusion rests on the proved state recurrence and an exact state repetition, not on extrapolating a numerical search. No floating-point arithmetic, random search, external solver, or conjectured period is involved. Conjecture 4.1 of arXiv:2504.07317v1, dated April 9, 2025, is refuted; Question 4.6 of arXiv:2504.07317v3, dated January 5, 2026, and credited in that revision to Ignacio García-Marco, is answered negatively. That was the latest source version located. No earlier resolution was located in the literature search, but that search is not an exhaustive priority determination, and priority has not been independently established.

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Provenance

This report merges two independently produced research reports on the same problem. The first, `ordinal-chomp-transition-at-two`, dated September 20, 2026 and prepared with ChatGPT in response to a research request, proves the transition for $\langle 4, 5, 6 \rangle$ as its headline result and carries a second certificate for $\langle 4, 6, 9 \rangle$. The second, `ordinal-chomp-winner-stability`, dated September 19, 2026 and prepared for Vladimir Reshetnikov, treats $\langle 4, 6, 9 \rangle$ alone and works throughout in the poisoned-board convention. Both dates and both derivations are recorded here so that the earlier work on the second semigroup is not silently redated.

The theorem the two reports share, $\text{ch}(\{4, 6, 9\}) = 2$ with least winning opening $\omega 4 + 4$, was proved by the same argument in both: the same order decomposition, the same three-block detector, and a finite certificate over the same seventeen gap ideals. A cell-by-cell comparison confirms that the two certificates record the same 253 rows, agreeing in all 4301 truncated entries and in all 4301 untruncated values once the column ordering [equation \(8.11\)](#) and the convention shift [equation \(8.7\)](#) are applied. There is one proof here, run on two semigroups, and it is given once.

Material that genuinely differed between the two sources has been kept from both and is marked where it appears. The principal such items are the shift corollary [theorem 3.2](#), the free-standing detector [theorem 3.3](#) with [theorem 3.4](#), the alternative criterion [theorem 3.6](#), the block-by-block reply catalogue of [section 8.10](#) beside the invariant argument of [section 7](#), the finite gadget of [section 8.9](#), the complete $\langle 4, 6, 9 \rangle$ certificate table in [section B](#), and the regression suite described in [section 8.8](#).

1 The problem and the result

A numerical semigroup is an additive submonoid $S \subseteq \mathbb{N}$, containing 0, whose complement in $\mathbb{N}$ is finite. Here $\mathbb{N}$ includes 0. Its associated partial order is

$$x \leq_S y \iff y - x \in S. \quad (1.1)$$

In Chomp, a move deletes the chosen element and all elements above it. The least element 0 is poisonous: a player forced to take it loses. Equivalently, remove 0 from the board at the outset and use normal play, under which a player with no legal move loses.

For an ordinal σ , form the coefficient extension

$$S^\sigma = \{\omega^{\xi_1} s_1 + \dots + \omega^{\xi_k} s_k : \sigma > \xi_1 > \dots > \xi_k, \quad s_i \in S \setminus \{0\}\} \cup \{0\}. \quad (1.2)$$

It carries the order defined by ordinary ordinal subtraction: $\alpha \leq_{S^\sigma} \beta$ if $\alpha \leq \beta$ and $\beta - \alpha \in S^\sigma$. The subtraction is characterized by $\alpha + (\beta - \alpha) = \beta$. In particular, $S^1 = S$, while S^2 is a lexicographic product of two copies of the semigroup poset, not their coordinatewise product.

Rivero Herrera’s Question 4.6 asks whether the transition exponent for these generated ordinal posets is always the smallest possible one [\[1\]](#); the question is credited in that revision to Ignacio García-Marco. For finite natural-number generators, that assertion would mean that whenever some extension is a first-player win, the first extension already is. The earlier version states the natural-number case as Conjecture 4.1 [\[2\]](#): it asks whether a second-player win on the first extension necessarily remains a second-player win on all ordinal extensions. We call that assertion *winner stability*, and we refute it. We address precisely the natural-number case; a generic finite-poset example would not suffice, since the newer paper already distinguishes the unrestricted poset problem from its algebraically generated specialization.

Write $\mathcal{P}$ for a losing position for the player to move and $\mathcal{N}$ for a winning one. For a finite set of positive integer generators Γ , define

$$\text{ch}(\Gamma) = \min\{\sigma \geq 1 : \text{Chomp on } \langle \Gamma \rangle^\sigma \text{ is } \mathcal{N}\},$$

provided the set is nonempty. We do not assign this minimum when all extensions are $\mathcal{P}$.

Theorem 1.1 (Main counterexample). *Let $\Gamma = \{4, 5, 6\}$ and $S = \langle \Gamma \rangle$. Then*

$$\text{Chomp on } S^\sigma \text{ is } \begin{cases} \mathcal{P}, & \sigma = 0, 1, \\ \mathcal{N}, & \sigma \geq 2. \end{cases}$$

For every $\sigma \geq 2$, the least winning first move, in the ordinary ordinal order, is

$$\boxed{\omega \cdot 4 + 4}. \tag{1.3}$$

In particular, $\text{ch}(\{4, 5, 6\}) = 2$.

The answer to the minimal-transition question is therefore negative already for three small integer generators, and winner stability fails. No uncountable ordinal, large-cardinal assumption, or change to the game rules is needed. The same architecture, carried out over a larger gap poset, gives a second and independent counterexample.

Theorem 1.2 (Second counterexample). *Let $\Gamma' = \{4, 6, 9\}$ and $S' = \langle \Gamma' \rangle$.*

- (i) *Chomp on S'^σ is $\mathcal{P}$ for $\sigma = 0$ and $\sigma = 1$; in particular Chomp on S' itself is a second-player win.*
- (ii) *Chomp on S'^2 is $\mathcal{N}$. The move $\alpha = \omega \cdot 4 + 4$ leaves a position of Grundy value 0.*
- (iii) *The same move wins on S'^σ for every $\sigma \geq 2$, and in each such extension it is the least winning first move in the ordinary ordinal order. Consequently $\text{ch}(\{4, 6, 9\}) = 2$.*

Part (i) is elementary and uses only the symmetry of this semigroup. Part (ii) is the substantive calculation: it rests on a finite certificate for a low-value exclusion, not on a simulation of an infinite board. Part (iii) holds because the chosen move removes every newly added higher-order block. [Section 8](#) proves all three parts. The second semigroup is stated here as a theorem in its own right, rather than as a corollary, because one of the two source reports had it as its headline result.

1.1 What is being claimed

The assertion is a counterexample within the numerical-semigroup family, not merely within arbitrary pointed partially ordered sets. Winner changes for arbitrary finite pointed posets are already exhibited in [1, Example 3.2], and those examples do not answer the semigroup question. Conversely, neither counterexample contradicts any asserted countable upper bound for a transition exponent: each realizes the very small nontrivial value 2.

The source check for this report found the question still present in the latest arXiv version, dated January 5, 2026, and found no later resolution. That is a statement about the literature located in this investigation, not a claim to have exhaustively searched unpublished work. The mathematical claim is supported by the complete reduction and the executable certificates below, independently of its eventual priority assessment.

1.2 Proof architecture

The proof has three logically separate parts. First, an elementary calculation gives the semigroup's gaps and an Apéry diamond. Second, Sprague–Grundy algebra turns one excluded option value into a winning move in the square. Third, a finite-state calculation establishes that exclusion for every possible first move in the base semigroup.

The infinite conclusion is not obtained by replacing a sufficiently large finite board by an infinite one. No such approximation is used. The only limit argument is the induction that follows an exactly repeated state of a deterministic recurrence.

1.3 Why the winner changes, and why this problem was tractable

The outcome of an isolated game records only whether its Grundy value is zero. A lexicographic substitution can see more: it can depend on which values occur *one move later*. This suggests looking for a second-player-winning semigroup whose finite remainders have a small missing Grundy value. In the examples below, the missing value is 2. That absence changes the value of a small three-block configuration from what the individual outcome classes alone would suggest.

In more detail, the poisoned base game loses because all of its first-move residuals have a maximum. In the square, that maximum becomes an entire upper copy of the semigroup. Its interaction with two incomparable lower copies detects a *gap in the option-value spectrum*: the lower pair has value 0, but its first two excluded option values are 0 and 3, not 0 and 2. Placing a value-1 block above the pair selects the second of these exclusions, and the independent Apéry diamond contributes another 3, so the two cancel. The additional ordinal coordinate is therefore not merely adding more moves to the same strategic situation. It packages the base game into an ordered compound that can observe more than its winner, or even than its single Grundy value. That is exactly why a winner-preservation conjecture can fail.

The decisive calculation is finite but not a bounded experiment. A numerical semigroup has only finitely many gaps. Those gaps encode all proper downsets by a moving boundary and finitely many possible tails. After retaining only the information needed to distinguish values 0, 1, and 2, the boundary evolves by a finite-memory recurrence. An exact repetition of its entire state proves the desired assertion for infinitely many Apéry parameters.

2 Orders, termination, and game conventions

2.1 Two games that must not be confused

For a rooted poset P with least element 0, let $C(P)$ denote its Chomp game, equivalently the normal-play game on $P \setminus \{0\}$. Let $T(P)$ denote the normal-play game on *all* of P , where 0 is playable and deletes the entire board. Throughout this paper, $g(Q)$ for a finite poset Q means the latter, ordinary normal-play Grundy value on all listed elements.

This distinction is essential inside a lexicographic product. The point $(0, 0)$ is the only poisoned point. For $a > 0$, the bottom $(a, 0)$ of a higher block is an ordinary legal move. Treating every block bottom as poisoned would produce a different game.

For a finite surviving board D containing 0, we call D itself the *unpoisoned* board and $D \setminus \{0\}$ its *poisoned* version. Both occur in the files shipped with this report, and [section 8.4](#) gives the

dictionary between them. Every value $g(\cdot)$ written in this article is the unpoisoned one, so that $g(C(P)) = g(P \setminus \{0\})$ and $g(T(P)) = g(P)$.

The study of this game, including the symmetric-semigroup second-player result and the finite-boundary representation used in [section 5](#), originates in García-Marco and Knauer [3]. We prove the particular facts needed here rather than treating their general theorems as black boxes.

For a terminating impartial game G , put

$$\mathcal{A}(G) = \{g(G') : G \longrightarrow G'\}, \quad g(G) = \text{mex } \mathcal{A}(G). \quad (2.1)$$

The mex is the least excluded ordinal; finite positions have finite values. A position is $\mathcal{P}$ exactly when its value is zero. Infinite boards in this paper are terminating but need not have a uniform finite bound on the number of moves. We do not assume finite branching.

2.2 The lexicographic order at exponent two

Every member of S^2 is uniquely $\omega a + b$ with $a, b \in S$. If $a < c$, then

$$(\omega c + d) - (\omega a + b) = \omega(c - a) + d.$$

If $a = c$ and $b \leq d$, the difference is $d - b$. It follows that

$$(a, b) \leq_{S^2} (c, d) \iff (a <_S c) \text{ or } (a = c \text{ and } b \leq_S d). \quad (2.2)$$

Here $a <_S c$ means $c - a \in S \setminus \{0\}$. Thus unequal comparable first coordinates make *every* element of the lower block lie below *every* element of the upper block.

The algebraic construction from natural sums and products of integer generators gives exactly [equation \(1.2\)](#): each Cantor coefficient is an ordinary nonnegative integer combination of the generators. The order, however, uses ordinary ordinal subtraction, not coefficientwise subtraction under the natural sum. Our convention [equation \(2.2\)](#) explicitly supplies the reflexive case and uses a strict comparison of unequal first coordinates. This also explains why the winning opening must be written $\omega \cdot 4 + 4$ and not $4 \cdot \omega + 4$: those are different ordinals, and only the first is a member of S^2 .

2.3 Why no infinite play occurs

Let F be the largest gap of a numerical semigroup $S \neq \mathbb{N}$. If $x < y$ and $y - x > F$, then $x <_S y$. Therefore an antichain in S is bounded in an interval of length F , and it is finite. A descending chain for $\leq_S$ is also a descending chain of natural numbers. Hence S is a well partial order.

A lexicographic product of two well partial orders with least elements is again a well partial order. Indeed, the distinct first coordinates of an antichain must themselves be an antichain, and a fixed fiber contains only a finite antichain. A descending sequence is excluded by first stabilizing the first coordinate and then the second, or, in our ordinal realization, simply by the ordinary ordinal order.

The same reasoning proves by transfinite induction that every S^σ is a well partial order. At a limit stage, two nonzero ordinals with different leading exponents are comparable: their highest differing coordinate compares 0 with a positive member of S . An antichain therefore has a fixed leading exponent and belongs to a smaller successor stage, where the induction hypothesis applies. The point 0 creates no exception.

In any Chomp play, the chosen sequence $x_0, x_1, \dots$ satisfies $x_i \not\leq x_j$ for $i < j$, because an earlier move would otherwise have deleted the later one. The well-partial-order property excludes such an infinite sequence. This justifies the terminating-game recursions used below.

For the two semigroups treated here, a reader can also check termination by hand, without the general theory. Every play of either game on S terminates because its first move either empties the board or leaves a finite Apéry set. After the prescribed move in the square only finitely many blocks remain, and once a block is touched it becomes finite. Hence all the Grundy values used in the main argument are well defined, even though some of the games involved have infinitely many options.

3 The small amount of game algebra needed

3.1 Disjoint and ordinal sums

The disjoint sum $G + H$ lets a player move in one component and leave the other unchanged. Its value is

$$g(G + H) = g(G) \oplus_N g(H). \quad (3.1)$$

For finite values, $\oplus_N$ is bitwise exclusive-or, not Hessenberg natural sum. The normal-play theorem and the ordinal-sum formula below are standard; a convenient treatment is [4, Sections 2 and 4, especially Theorem 2.6]. We include the part of the proof that controls the counterexample.

The ordinal sum $G : H$ has G below H . A move in H preserves G ; a move in G discards all of H . Therefore its options are G' and $G : H'$. If A is a set of ordinals, define its excluded-ordinal enumeration recursively by

$$E_A(\eta) = \text{mex}(A \cup \{E_A(\xi) : \xi < \eta\}). \quad (3.2)$$

It lists the complement of A in increasing order, beginning with index 0.

Lemma 3.1 (Ordinal-sum formula). *For terminating impartial games,*

$$g(G : H) = E_{\mathcal{A}(G)}(g(H)).$$

In particular, if $g(H) = 0$, then $g(G : H) = g(G)$.

Proof. Fix G and argue by induction on the game tree of H . Moves in G have values in $A = \mathcal{A}(G)$; by induction moves in H have values $E_A(g(H'))$. Put $\eta = g(H)$. No option of H has value η , while every $\xi < \eta$ occurs as an option value. Thus $E_A(\eta)$ is absent from the option values of $G : H$. Every ordinal below it is either in A , or equals $E_A(\xi)$ for some $\xi < \eta$, and is present. Taking mex proves the formula. The last assertion is $E_A(0) = \text{mex } A$. $\square$

A first, elementary consequence records what a playable minimum does. It is quoted repeatedly below, and it is what makes the poisoned and unpoisoned conventions interchangeable.

Corollary 3.2 (Adding a legal minimum). *If D is a finite poset with least element 0, then*

$$g(D) = 1 + g(D \setminus \{0\}). \quad (3.3)$$

Proof. Write $D = \mathbf{1} : (D \setminus \{0\})$, where $\mathbf{1}$ is the one-point game: playing 0 in D deletes everything, and every other move leaves 0 in place below the remainder. Since $\mathcal{A}(\mathbf{1}) = \{0\}$, the n -th ordinal excluded from $\mathcal{A}(\mathbf{1})$ is $n + 1$ for finite n . Now apply [theorem 3.1](#). $\square$

The option spectrum of a disjoint sum also has a simple description:

$$\mathcal{A}(G + H) = \{a \oplus_N g(H) : a \in \mathcal{A}(G)\} \cup \{g(G) \oplus_N b : b \in \mathcal{A}(H)\}. \quad (3.4)$$

This follows immediately by considering which component was changed.

3.2 A low-value detector

The whole mechanism of this paper is concentrated in the following lemma, which mentions no posets at all. Isolating it separates the nimber content from the order geometry, and both criteria below cite it.

Lemma 3.3 (The low-value detector). *Let G be a terminating impartial game with $g(G) = 1$, all values in $B = \mathcal{A}(G)$ finite, and*

$$B \cap \{0, 1, 2, 3\} = \{0, 3\}. \quad (3.5)$$

Then

$$g((G + G) : G) = 3. \quad (3.6)$$

Proof. Put $U = G + G$. Then $g(U) = 1 \oplus_N 1 = 0$, and by [equation \(3.4\)](#),

$$C := \mathcal{A}(U) = \{b \oplus_N 1 : b \in B\}.$$

Exclusive-or with 1 exchanges 0 with 1 and 2 with 3, so [equation \(3.5\)](#) gives

$$1, 2 \in C, \quad 0, 3 \notin C.$$

These four assertions correspond respectively to $0, 3 \in B$ and $1, 2 \notin B$; option values at least 4 cannot affect them. Hence the first two ordinals excluded from C are $E_C(0) = 0$ and $E_C(1) = 3$. Since $g(G) = 1$, [theorem 3.1](#) gives $g((G + G) : G) = E_C(1) = 3$. $\square$

Remark 3.4 (Why a single nimber is not enough). Replacing *every* occurrence of G by a singleton would be wrong. A singleton 1 also has Grundy value 1, yet

$$g((1 + 1) : 1) = 2,$$

not 3: here $\mathcal{A}(1 + 1) = \{1\}$, whose first two excluded values are 0 and 2. The lower components of an ordinal sum therefore contribute their option *spectrum*, not merely its mex, while the upper component may be summarized by its Grundy value. The distinction is systematic in the ordered-join substitution theory of [4, Section 4].

3.3 A four-point-remainder criterion

Let P be a rooted well partial order. Abbreviate $C = C(P)$ and $T = T(P)$. A finite diamond D is the four-element poset with bottom, two incomparable middle elements, and top. All its points are playable, and $g(D) = 3$: deleting the bottom, a middle point, or the top gives values 0, 2, or 1, respectively.

Proposition 3.5 (Spectral mechanism). *Suppose $a \in P \setminus \{0\}$ satisfies the following conditions. The remainder $P \setminus \uparrow a$ is a diamond $\{0, p, q, r\}$, and a is incomparable with p, q, r . Moreover,*

$$g(C) = 0, \quad \mathcal{A}(T) \cap \{0, 1, 2, 3\} = \{0, 3\}. \quad (3.7)$$

Then (a, a) is a winning opening in Chomp on $P \text{ lex } P$.

Proof. Since T is a playable singleton below C , [theorem 3.1](#) gives $g(T) = 1$. The hypothesis [equation \(3.7\)](#) is exactly [equation \(3.5\)](#) for $G = T$, and all option values of T are finite because every option is a finite poset. Therefore [theorem 3.3](#) applies and

$$g((T + T) : T) = 3. \quad (3.8)$$

Explicitly, with $U = T + T$ and $A = \mathcal{A}(T)$ we have $g(U) = 0$ and

$$\mathcal{A}(U) \cap \{0, 1, 2, 3\} = \{1, 2\}, \quad (3.9)$$

whose first two excluded ordinals are 0 and 3.

After the move (a, a) , the first-coordinate blocks indexed by $0, p, q, r$ remain complete, and the block indexed by a becomes $D = P \setminus \uparrow a$. All other blocks disappear. The poisoned point lies only in block 0, so that block contributes C . The three blocks at p, q, r contribute $(T + T) : T$, while the block at a is incomparable with all three. The remaining game is exactly

$$C : (D + ((T + T) : T)). \quad (3.10)$$

The game in parentheses has value $3 \oplus_{\mathbb{N}} 3 = 0$. Its ordinal sum above C therefore has value $g(C) = 0$. The opening leaves a $\mathcal{P}$ -position and wins. $\square$

The condition excluding 2 from the option spectrum is genuinely used. Knowing only $g(T) = 1$ excludes 1, but does not determine whether 2 occurs. If 2 occurred, the second excluded value of $\mathcal{A}(T + T)$ would no longer be forced to equal 3. This is the information lost by an argument based only on the outcome of an individual block. [Theorem 3.4](#) makes the complementary point with an explicit counter-computation.

3.4 An alternative criterion, with weaker hypotheses

The second of the two source reports isolated a variant of [theorem 3.5](#) that assumes less. We keep both. The version below buys two things: incomparability is *derived* from atomicity rather than assumed, and the low-spectrum hypothesis shrinks to the single exclusion of the value 2, because $0 \in \mathcal{A}(G)$ now comes from the minimum move and $3 \in \mathcal{A}(G)$ from the diamond option.

Proposition 3.6 (Diamond-atom criterion). *Let P be a pointed poset whose normal and poisoned games terminate, and write $G = \mathsf{T}(P)$. Suppose that*

- (a) $g(P \setminus \{0\}) = 0$ and $g(G) = 1$;
- (b) P has an atom a for which $P \setminus \uparrow a$ is the four-point diamond;
- (c) $2 \notin \mathcal{A}(G)$, and all values in $\mathcal{A}(G)$ are finite.

Then (a, a) is a winning first move in Chomp on $P \text{ lex } P$.

Proof. The minimum move gives $0 \in \mathcal{A}(G)$; the value $g(G) = 1$ excludes 1 from $\mathcal{A}(G)$; the option at a is the diamond, of value 3, so $3 \in \mathcal{A}(G)$. With (c) this is [equation \(3.5\)](#) for G . The atom a is incomparable with every positive point of $P \setminus \uparrow a$, since it is neither above a positive predecessor, being an atom, nor below a surviving point, by the definition of $\uparrow a$. The residual decomposition after (a, a) is therefore exactly [equation \(3.10\)](#), and [theorem 3.3](#) finishes the computation as in [theorem 3.5](#). $\square$

The hypotheses of [theorem 3.6](#) are strictly weaker than those of [theorem 3.5](#) in the situations used here, and imply them: given (a), (b) and (c), the incomparability required by [theorem 3.5](#) follows from atomicity, and $\mathcal{A}(T) \cap \{0, 1, 2, 3\} = \{0, 3\}$ follows as in the proof just given. The converse does not hold, since [theorem 3.5](#) does not require a to be an atom. We nevertheless retain [theorem 3.5](#) because it is the form applied verbatim in [section 4](#), where the incomparabilities are checked directly from the gap list and no atom argument is needed.

4 The semigroup $\langle 4, 5, 6 \rangle$

4.1 Symmetry and the losing first level

Let $S = \langle 4, 5, 6 \rangle$. The four consecutive integers 8, 9, 10, 11 lie in S , and adding 4 proves that every integer at least 8 lies in S . Direct inspection below 8 gives

$$S = \mathbb{N} \setminus \{1, 2, 3, 7\}, \quad F = 7. \quad (4.1)$$

For $0 \leq n \leq 7$, exactly one of n and $7 - n$ belongs to S . Thus S is symmetric.

For $b \in S \setminus \{0\}$, define the Apéry remainder

$$R_b = \text{Ap}(S, b) = S \setminus (b + S). \quad (4.2)$$

It is exactly the board remaining after the move b in S , before removing the poisoned 0. It is finite: an element larger than $b + F$ cannot remain. We also set $R_0 = \text{Ap}(S, 0) = \emptyset$ when using normal play, which is the value of the move at 0 in $T(S)$.

Lemma 4.1 (The symmetric-semigroup argument). *Every R_b for $b > 0$ has greatest element $b + 7$. Consequently $C(S)$ is a $\mathcal{P}$ -position.*

Proof. The element $b + 7$ belongs to S and its difference from b is the gap 7, so it belongs to R_b . If $x \in R_b$ and $x < b$, then $b + 7 - x > 7$ belongs to S . If $x \geq b$, then $x - b$ is a gap, and symmetry gives $7 - (x - b) \in S$. In both cases $x \leq b + 7$.

A finite normal-play poset Q with a greatest element m is $\mathcal{N}$. To see this, delete m . If the result is $\mathcal{P}$, that was a winning move. If not, it has a winning move z ; playing z in the original poset deletes m as well and gives the same $\mathcal{P}$ -remainder. Apply this to $Q = R_b \setminus \{0\}$, which still has the nonzero greatest element $b + 7$. Every legal first move in $C(S)$ thus leaves an $\mathcal{N}$ -position, proving the claim. $\square$

This reproduces, in the present small case, the symmetric-semigroup result of García-Marco and Knauer [3, Theorem 2.4]. The extra fact we need is stronger than that outcome statement.

Write $C = C(S)$ and $T = T(S)$. Since R_b for $b > 0$ is a finite poset with least element 0, [theorem 3.2](#) gives

$$g(R_b) = 1 + g(R_b \setminus \{0\}). \quad (4.3)$$

In particular, the preceding lemma excludes value 1 for R_b . Also

$$g(C) = 0, \quad g(T) = 1, \quad \mathcal{A}(T) = \{0\} \cup \{g(R_b) : b \in S \setminus \{0\}\}. \quad (4.4)$$

4.2 The diamond and the remaining spectral assertion

A direct calculation gives

$$R_4 = \{0, 5, 6, 11\}. \quad (4.5)$$

Its middle points are incomparable because $6 - 5 = 1 \notin S$; its top dominates them because $11 - 5 = 6$ and $11 - 6 = 5$ are in S . Thus it is a diamond D , and $g(D) = 3$. Moreover, 4 is incomparable with 5, 6, and 11: their differences from 4 are the gaps 1, 2, and 7.

Lemma 4.2 (Certified spectral gap). *For every $b \in \langle 4, 5, 6 \rangle \setminus \{0\}$,*

$$g(R_b) \geq 3. \quad (4.6)$$

The proof occupies sections 5 and 6. Given this lemma, equation (4.4) has low spectrum exactly $\{0, 3\}$, and theorem 3.5 applies with $a = 4$, $p = 5$, $q = 6$, $r = 11$. The block decomposition after $\omega 4 + 4$ is displayed below.

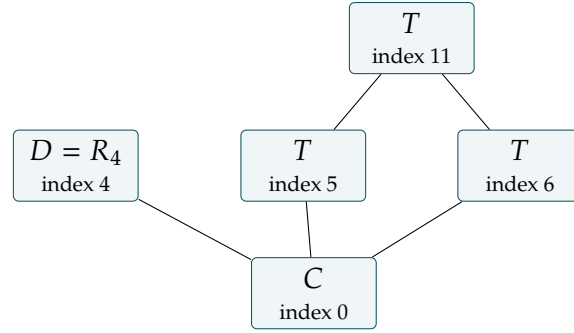


Figure 1: The remaining blocks, with lower boxes below upper boxes. Only the global point $(0, 0)$ was poisoned. The block at index 4 is detached from the three blocks at indices 5, 6, and 11.

4.3 The exact transition and the least winning opening

The move $\omega 4 + 4$ leaves the game

$$C : (D + ((T + T) : T))$$

and hence is winning, once theorem 4.2 has been established. No positive *finite* opening is winning: such a move deletes every ordinal with a positive higher coefficient and leaves just $R_b \setminus \{0\}$, which is $\mathcal{N}$ by theorem 4.1.

The only remaining legal opening below $\omega 4 + 4$ is $\omega 4$. After it, the block at index 4 is absent. The opponent can play $\omega 11$, deleting the top block and leaving

$$C : (T + T),$$

which is $\mathcal{P}$, since $g(C) = 0$ and $g(T + T) = 0$. Thus $\omega 4$ is not a winning first move either. There are no intermediate legal coefficients 1, 2, or 3. This proves minimality of equation (1.3).

Finally, in S^σ with $\sigma > 2$, every point whose leading exponent is at least 2 lies above $\omega 4 + 4$: subtracting the latter leaves that point itself. The opening therefore deletes every such point and leaves exactly the same board as it does in S^2 . This proves all assertions of theorem 1.1, conditional only on the certified spectral lemma proved next.

5 A finite-window theorem for low Grundy values

We now give a general reduction, rather than postulating periodicity for one numerical sequence. Let $S \neq \mathbb{N}$ be an arbitrary numerical semigroup, $\mathcal{H} = \mathbb{N} \setminus S$ its gap set, and $F = \max \mathcal{H}$. The moving-boundary encoding of downsets follows the locality idea used in [3, Section 6]; retaining several low Grundy values instead of just the outcome gives the recurrence developed here. No claim of independent priority is made for this general method.

5.1 Encoding all proper downsets

Give $\mathcal{H}$ the order induced by semigroup differences, and let $\mathcal{J}$ be the finite set of its downsets. For $x > F$ and $C \in \mathcal{J}$, set

$$P(x, C) = (S \cap [0, x)) \cup (x + C). \quad (5.1)$$

All these positions include a *playable* 0.

Lemma 5.1. *The finite downsets of S whose least missing integer of S is $x > F$ are exactly the positions $P(x, C)$ with $C \in \mathcal{J}$. In particular, $R_x = P(x, \mathcal{H})$ for $x > F$.*

Proof. If P is such a downset and $z \in P$ with $z \geq x$, the difference $z - x$ cannot belong to S , or the missing point x would lie below z . Thus $z = x + g$ for a gap g . If g is included and $g - h \in S$ for another gap h , then $x + h \leq_S x + g$, so h is included too. The selected gaps form $C \in \mathcal{J}$.

Conversely, let C be a downset of the gap order. Suppose $z \leq_S x + g$ with $g \in C$. If $z < x$, it is in the prescribed prefix. If $z \geq x$, write $z = x + h$. The number h cannot lie in S , because $g - h \in S$ would then imply $g \in S$. Hence $h \in \mathcal{H}$, and the downset property puts h in C . This proves that $P(x, C)$ is a downset. Its least missing member is x , since 0 is not a gap. The last assertion follows directly from the definition of R_x . $\square$

This description *refines* the finite-gap encoding of [3, Section 6]: beyond the conductor, the admissible tails are exactly the gap ideals, not arbitrary subsets of the gap set. That is what makes the number of columns finite and small, seventeen rather than sixty-four in the larger example below.

5.2 Truncation without approximation

Fix $K \geq 0$. Define

$$\tau_K(n) = \begin{cases} n, & 0 \leq n \leq K, \\ *, & n > K. \end{cases}$$

For a set of these symbols, $\text{mex}_{\leq K}$ returns its least absent member of $\{0, \dots, K\}$, or $*$ if every one is present. The symbol $*$ is not a number. Then

$$\tau_K(\text{mex } B) = \text{mex}_{\leq K} \{\tau_K(b) : b \in B\} \quad (5.2)$$

for every finite set of option values B . Indeed, deciding whether the mex is a specified number at most K requires knowing only which option values up to K occur. Therefore computing the truncated values recursively is *exact*, not a numerical approximation to larger values.

Put

$$v_x(C) = \tau_K(g(P(x, C))).$$

The proof uses $K = 2$, but the general statement will be useful for understanding why a finite certificate is possible.

5.3 Every move is either far, near, or in the tail

For a tail move at $x + c$, $c \in C$, the new tail is

$$C_c = \{g \in C : g - c \notin S\}. \quad (5.3)$$

Negative integers are understood not to belong to S . In particular, $c \notin C_c$, so this is a proper smaller tail.

For a prefix move at $y = x - d$, where $1 \leq d \leq F$ and $x > 2F$, define

$$\Delta_d(C) = \{g \in \mathcal{H} : g < d \text{ or } g - d \in C\}. \quad (5.4)$$

The new position is $P(x - d, \Delta_d(C))$. To check this, a possible surviving tail point is $y + g$ with $g \in \mathcal{H}$. If $g < d$, it was in the old full prefix; if $g = d$, it was the missing point x ; if $g > d$, it was present exactly when $g - d \in C$. All nearby prefix choices are legal because $x - d > F$.

If the prefix move is farther away, $y < x - F$, the complete Apéry remainder R_y already lies below x , since its largest possible element is $y + F < x$. The resulting position is exactly R_y , independently of the old tail. This includes the normal-play move $y = 0$, whose remainder is empty.

Both C_c and $\Delta_d(C)$ belong to $\mathcal{J}$. They encode downsets resulting from legal moves; alternatively their closure can be checked directly from [equations \(5.3\) and \(5.4\)](#).

These three cases are identities of *actual finite posets*. They do not depend on any assumption about which player wins, or on any Grundy value at all. That independence is what licenses the separate literal residual-identity check performed in code and reported in [section 8.8](#).

Define the accumulated far spectrum

$$B_x = \{g(R_y) : y \in S, 0 \leq y < x - F\} \cap \{0, \dots, K\}. \quad (5.5)$$

Combining the three kinds of moves gives the exact recurrence

$$v_x(C) = \text{mex}_{\leq K} \left(B_x \cup \{v_{x-d}(\Delta_d(C)) : 1 \leq d \leq F\} \cup \{v_x(C_c) : c \in C\} \right), \quad x > 2F. \quad (5.6)$$

For a fixed x , evaluate smaller tails first. The same-row terms in [equation \(5.6\)](#) are therefore already known. The far spectrum updates by

$$B_{x+1} = B_x \cup (\{v_{x-F}(\mathcal{H})\} \cap \{0, \dots, K\}). \quad (5.7)$$

5.4 Why a repeated state is a proof for all parameters

Theorem 5.2 (Finite-window principle). *Let $S \neq \mathbb{N}$ be a numerical semigroup and fix $K \geq 0$. The vector-valued sequence $(v_x(C))_{C \in \mathcal{J}}$ is ultimately periodic. Its eventual behavior is effectively determined by the finite initial data*

$$\{v_x(C) : F < x \leq 2F, C \in \mathcal{J}\}, \quad B_{2F+1}.$$

In particular, for any fixed finite k , it is decidable whether $g(R_b) = k$ for some positive $b \in S$.

Proof. The complete state before computing row x is

$$\mathcal{S}_x = (B_x, v_{x-F}, v_{x-F+1}, \dots, v_{x-1}), \quad v_t = (v_t(C))_{C \in \mathcal{J}}. \quad (5.8)$$

Equations (5.6) and (5.7) determine its successor without using x itself. There are at most

$$2^{K+1}(K+2)^{F|\mathcal{J}|} \quad (5.9)$$

possible states: the far spectrum is a subset of $K+1$ labels, and there are $F|\mathcal{J}|$ row entries, each with $K+2$ possibilities. Some states are impossible, but this finite upper bound suffices. A repeated complete state has the same future by induction; equivalently, a repetition occurs among any $2^{K+1}(K+2)^{F|\mathcal{J}|} + 1$ consecutive states.¹

All seed positions are finite and explicitly given. The initial far set is computed from the finitely many R_y with $y \leq F$, including R_0 . Hence the finite orbit can be computed until a repetition occurs. To decide occurrence of a value k , use $K = k$, check the exceptional parameters $b \leq F$, and inspect the finite preperiod and period of the full-tail column. $\square$

When $S = \mathbb{N}$, R_b is a chain of b points and has value b , so the final decidability assertion is immediate as well.

Remark 5.3 (What is and is not periodic). The theorem concerns every *fixed bounded range* of Grundy values, one range at a time. It does not assert that the *untruncated* Grundy sequence is periodic, and the separate cutoffs may have *different preperiods and different periods*. Nor does it decide the winner at every ordinal exponent for every semigroup. It proves exactly the finite-number information used here.

6 The nine-shape certificate

6.1 Shapes and initialization

For $S = \langle 4, 5, 6 \rangle$, the gap poset has the three incomparable points 1, 2, and 3, all below 7. Its downsets are the eight subsets of $\{1, 2, 3\}$ and the full set $\mathcal{H} = \{1, 2, 3, 7\}$. Encode a subset by

$$m(C) = \mathbf{1}_{1 \in C} + 2\mathbf{1}_{2 \in C} + 4\mathbf{1}_{3 \in C} + 8\mathbf{1}_{7 \in C}.$$

The shape order in every table and certificate is

$$0, 1, 2, 3, 4, 5, 6, 7, 15. \quad (6.1)$$

We take $K = 2$. In the machine-readable files, digit 3 represents *, meaning a value at least 3. In the mathematical tables, we print the saturation symbol instead.

The seed rows are $x = 8, \dots, 14$. They are computed directly from equations (2.1) and (5.1), without using the moving-boundary recurrence. The exceptional positive parameters are $b = 4, 5, 6$. Their exact normal-play values are

$$g(R_4) = 3, \quad g(R_5) = 5, \quad g(R_6) = 5. \quad (6.2)$$

¹The second source report states the same bound in the poisoned parametrization, as $N_k = 2^k(k+1)^{Fr}$ with $r = |\mathcal{J}|$ and cutoff k , and phrases the conclusion as “a repetition occurs among any $N_k + 1$ consecutive states”. The two agree under $k = K + 1$, since the poisoned labels $0, \dots, k - 1$ together with saturation give $k + 1 = K + 2$ possibilities per entry and $2^k = 2^{K+1}$ prefix sets.

Therefore $B_{15} = \{0\}$. Rows 15, $\dots$, 50 are then obtained from [equations \(5.6\)](#) and [\(5.7\)](#).

For complete transparency, the nearby-tail map is the following small table. Entries are masks, not Grundy values.

$m(C)$	$d = 1$	2	3	4	5	6	7
0	0	1	3	7	7	7	7
1	2	5	3	7	7	15	7
2	4	1	3	7	15	7	7
3	6	5	3	7	15	15	7
4	0	1	3	15	7	7	7
5	2	5	3	15	7	15	7
6	4	1	3	15	15	7	7
7	6	5	3	15	15	15	7
15	6	5	3	15	15	15	7

For masks below 8, a tail move simply deletes its chosen bit. For the full mask 15, moves at gaps 1, 2, 3, and 7 give masks 6, 5, 3, and 7, respectively. These rules and the preceding table are an explicit finite specification of the entire transition.

6.2 The certificate and its closure

Proposition 6.1 (Finite verification). *The 43 rows $v_8, \dots, v_{50}$ in [section A](#) satisfy the seed calculations and the exact recurrence. They have the following properties:*

$$v_x(\mathcal{H}) = * \quad (8 \leq x \leq 50), \quad (6.3)$$

$$B_x = \{0\} \quad (15 \leq x \leq 51), \quad (6.4)$$

$$(v_{36}, \dots, v_{42}) = (v_{44}, \dots, v_{50}). \quad (6.5)$$

Thus the complete states satisfy

$$\boxed{\mathcal{S}_{43} = \mathcal{S}_{51}}. \quad (6.6)$$

Finite verification procedure. For the seven seed rows, recursively delete each possible principal upper set from the finite position and take truncated mex. For subsequent rows, process the nine masks in nondecreasing cardinality, apply the displayed nearby-tail table, and apply the stated tail-cut rules. At each step update B_x by [equation \(5.7\)](#). These operations give the complete table in [section A](#). The full-tail column is always $*$, so the far set never gains 1 or 2. Direct comparison of the seven indicated pairs of rows proves [equation \(6.5\)](#). The corresponding far sets agree, so the comparison is of the *entire* state.

A self-contained executable version of this calculation is printed in [section C](#). A separate checker computes the untruncated Grundy value of every one of the 387 listed positions using bit masks and literal upper-set deletion. All comparisons passed. $\square$

Proof of [theorem 4.2](#). By [theorem 5.2](#) and [equation \(6.6\)](#), the complete state repeats with period 8, and the row vectors satisfy

$$v_{x+8} = v_x \quad (x \geq 36).$$

For $36 \leq x \leq 42$ this is the recorded seven-row equality; for $x \geq 43$ it follows inductively from the deterministic state transition. All full-tail values from row 8 onward are consequently $*$: they are

checked through row 50, and every later one is a repeated value. Since $R_x = P(x, \mathcal{H})$ for $x > 7$, all parameters above 7 have value at least 3. The only positive parameters below 8 are covered by [equation \(6.2\)](#). This proves the universal statement [equation \(4.6\)](#). $\square$

Notice what has *not* been assumed: it is not enough that the full-tail column stays nonzero for many terms, or that one row repeats. The certificate records seven consecutive complete row vectors *and* the accumulated far spectrum. Those are exactly the inputs required by the next transition.

6.3 A sample recurrence calculation

For an illustration, take row $x = 15$ and the empty tail $C = \emptyset$. The nearby masks for $d = 1, \dots, 7$ are

$$0, 1, 3, 7, 7, 7, 7.$$

Reading the corresponding entries in rows 14, 13, 12, 11, 10, 9, 8 gives

$$2, 2, 2, *, 1, 1, *.$$

Together with the far value 0, these contain 0, 1, and 2, so $v_{15}(\emptyset) = *$. There are no same-row tail moves in this example. The other eight entries of row 15 use the same rule, with smaller-tail values already evaluated.

This illustrates why the saturation symbol is sufficient. Values larger than 2 cannot change whether 0, 1, and 2 all occur among options; no rounded or estimated value enters the proof.

7 An explicit response procedure

The Grundy calculation proves the existence of a winning strategy. Here we make the responses constructive without asking for the Grundy value of an arbitrary infinite position. Every auxiliary computation in the procedure is on a finite poset.

We describe the strategy for the player who has just opened at $\omega 4 + 4$, and who now responds to the opponent's moves. Let $D = R_4$ and let T_1, T_2 be the two full blocks at indices 5 and 6. The cap is the block at index 11. Initially it is a full copy of T . The bottom block C remains below everything else.

7.1 Finite moves to specified smaller values

If a finite position has Grundy value h , then for every integer $k < h$ it has a move to value k . This is immediate from mex. To find such a move, compute the values of all finite options and choose any option with the desired value. The direct verifier's finite-game routine implements this computation.

The diamond has particularly simple moves:

$$D \longrightarrow 0 \text{ by playing } 0, \quad D \longrightarrow 1 \text{ by playing } 11, \quad D \longrightarrow 2 \text{ by playing } 5 \text{ or } 6.$$

A move at b in a full T leaves the finite position R_b . For $b = 0$ its value is 0; for $b > 0$ its value is at least 3.

7.2 The initial invariant

Maintain, when possible, the upper game

$$D + ((T_1 + T_2) : Z), \quad g(Z) = 1, \quad (7.1)$$

where D is still the intact diamond and Z is either the original full T or a finite cap. We call this phase I. Initially it holds. There are three locations for the opponent's move.

A move in the diamond. Its new value d is 0, 1, or 2. If $d = 0$, move in the cap to value 0; such a move exists because $g(Z) = 1$. If the cap is still T , play its bottom 0. The result is phase II below. If $d = 1$, play 0 in either lower full block. That deletes both that block and the cap, leaving a disjoint sum of two value-1 games. If $d = 2$, play 4 in either lower full block. Its new value is 3 and the other block has value 1, giving total xor $2 \oplus_N 3 \oplus_N 1 = 0$. The latter two cases enter phase III.

A move in the cap. The new cap W is finite, even when the old cap was T . Its value cannot be 1, since it is an option of a value-1 game. If $g(W) = 0$, delete the diamond by playing its bottom and enter phase II. If $g(W) \geq 2$, make a finite move in W to value 1, restoring phase I.

A move in a lower full block. The cap disappears, and the changed block becomes a finite R_b . If $b = 0$, its value is 0; change the diamond to value 1, canceling the other full block. If $g(R_b) = 3$, change the diamond to value 2, giving xor $2 \oplus_N 3 \oplus_N 1 = 0$. In the remaining case $g(R_b) \geq 4$, move inside R_b to value 2, again giving $3 \oplus_N 2 \oplus_N 1 = 0$. All these cases enter phase III. The spectral gap rules out precisely the missing troublesome case $g(R_b) = 2$.

7.3 The two continuation invariants

In **phase II**, the diamond is absent and the upper game is

$$(T_1 + T_2) : Z, \quad Z \text{ finite}, \quad g(Z) = 0.$$

After an opponent's move in the cap, its finite value is positive; move in it to value 0 again. After an opponent's move at b in a lower block, the cap disappears. For $b = 0$, simply delete the other full block at 0. For $b > 0$, the finite changed block has value at least 3, so move in it to value 1, canceling the other full block. This enters phase III. An empty cap causes no problem.

In **phase III**, the upper position is a disjoint sum of finite positions and possibly some full copies of T , with total xor zero. Every component value is a finite integer, and each full copy has value 1. After an opponent's move the xor becomes a nonzero integer z . Choose a component of value h for which $h \oplus_N z < h$, and move it to value $h \oplus_N z$. Such a component exists by the usual highest-bit argument for Nim. For a finite component, find the move by finite search. For a full component, $h = 1$, so the smaller target must be 0; playing its bottom realizes it. Thus no search through infinitely many moves is needed in the response calculation.

7.4 Moves in the bottom block

At every stage, a move by the opponent in the bottom C deletes all upper blocks. The remainder is the finite position $R_b \setminus \{0\}$ for some $b > 0$. Its value is positive by [theorem 4.1](#) (in fact at least 2

by the spectral lemma). Choose a move to value 0 and thereafter use ordinary finite-poset play, restoring value 0 after each opponent move.

These cases exhaust all legal opponent moves. They also show that all prescribed responses are legal. Each finite search terminates, and every play of the underlying poset game is finite. Therefore the opponent eventually has no legal move. This is a constructive strategy description, not merely a named winning opening. No efficient worst-case running-time claim is made for the finite game searches.

8 The second semigroup $\langle 4, 6, 9 \rangle$

This section proves [theorem 1.2](#). Everything in it is a second instance of the architecture already established, not a second proof: the game algebra of [section 3](#) and the finite-window theorem [theorem 5.2](#) are quoted, never reproved. What is genuinely new here is the larger gap poset, its seventeen-column certificate, and a block-by-block description of the winning responses that names actual ordinal moves.

8.1 Gaps, symmetry, and the losing first level

Lemma 8.1 (Gaps and symmetry). *For $S' = \langle 4, 6, 9 \rangle$,*

$$\mathcal{H}' = \mathbb{N} \setminus S' = \{1, 2, 3, 5, 7, 11\}, \quad (8.1)$$

the conductor is 12, and the largest gap is $F' = 11$. Moreover

$$n \in S' \iff 11 - n \notin S' \quad (0 \leq n \leq 11), \quad (8.2)$$

so S' is symmetric.

Proof. Below 12 the semigroup elements are exactly 0, 4, 6, 8, 9, 10. The four consecutive integers 12, 13, 14, 15 are generated, by

$$12 = 4 + 4 + 4, \quad 13 = 4 + 9, \quad 14 = 4 + 4 + 6, \quad 15 = 6 + 9,$$

and adding 4 to these gives every larger integer. This proves [equation \(8.1\)](#). For [equation \(8.2\)](#), inspect the six complementary pairs summing to 11: $\{0, 11\}$, $\{1, 10\}$, $\{2, 9\}$, $\{3, 8\}$, $\{4, 7\}$, $\{5, 6\}$. In each pair exactly one member lies in S' . $\square$

Write $C' = C(S')$ and $T' = T(S')$.

Lemma 8.2 (The losing first level). *For every positive $t \in S'$, the Apéry remainder $\text{Ap}(S', t)$ has greatest element $t + 11$. Consequently*

$$g(C') = 0, \quad g(T') = 1. \quad (8.3)$$

Proof. Put $M = t + 11$. Then $M \in S'$ while $M - t = 11 \notin S'$, so M lies in the Apéry set. Let z be any other member. If $z < t$, then $M - z > 11$, hence $M - z \in S'$. If $z \geq t$, the difference $z - t$ is a gap, and [equation \(8.2\)](#) gives $M - z = 11 - (z - t) \in S'$. In both cases $z \leq_{S'} M$.

A nonempty finite normal-play poset with a greatest element cannot have Grundy value 0; the argument is the one already given inside [theorem 4.1](#). Applying it to $\text{Ap}(S', t) \setminus \{0\}$, whose greatest

element is still M , shows that every option of C' is nonzero, so $g(C') = 0$. For T' , the move at 0 leaves the empty game, of value 0, while every positive move t leaves the finite board $\text{Ap}(S', t)$, whose value is at least 2 by [theorem 3.2](#) together with the preceding sentence. The mex is therefore $g(T') = 1$. $\square$

This reproves, for this particular semigroup, the symmetric-semigroup result of [3, Theorem 2.4]. It does *not* establish the additional exclusion of the value 1 from the poisoned Apéry boards, equivalently of the value 2 from the unpoisoned ones. Symmetry gives an outcome statement only; the stronger exclusion is precisely the certified lemma of [section 8.7](#).

8.2 The Apéry diamond for $\langle 4, 6, 9 \rangle$

Since 4 is the least positive element of S' , it is an atom of the semigroup order. Directly from [equation \(8.1\)](#),

$$D' = \text{Ap}(S', 4) = \{0, 6, 9, 15\}, \quad (8.4)$$

whose only strict cover relations are $0 < 6 < 15$ and $0 < 9 < 15$. Its middle points are incomparable because $9 - 6 = 3 \notin S'$; both lie below 15 because $15 - 6 = 9$ and $15 - 9 = 6$ belong to S' . Thus D' is a diamond. Its unpoisoned options are the following, with the residual shapes named.

Move	Remaining value	Remaining shape
0	0	Empty
6	2	The chain $0 < 9$
9	2	The chain $0 < 6$
15	1	A minimum below two incomparable points

Hence

$$g(D') = 3, \quad g(D' \setminus \{0\}) = 2, \quad (8.5)$$

the second equality by [theorem 3.2](#). The poisoned value 2 is the single most frequently quoted number in the ported certificate files, and the cleanest instance of the shift corollary.

Finally, 4 is incomparable with 6, 9, and 15, because their differences from 4 are the gaps 2, 5, and 11.

8.3 What remains after $\omega 4 + 4$

Select $(4, 4)$ in the lexicographic square. The surviving outer coefficients are 0, 4, 6, 9, 15; every other outer coefficient is removed because it lies strictly above 4 in the semigroup order. At outer coefficient 4 only the inner diamond D' remains, and at 0, 6, 9, 15 the whole inner semigroup remains.

Among the surviving positive outer coefficients, 4 is incomparable with 6, 9, and 15, as just noted; 6 and 9 are incomparable with each other and both lie below 15. The global poisoned point is $(0, 0)$, so the outer-0 block contributes C' and not T' , and every element of that block lies below all other surviving blocks. The residual game is therefore exactly

$$R' = C' : \left(D' + ((T' + T') : T') \right). \quad (8.6)$$

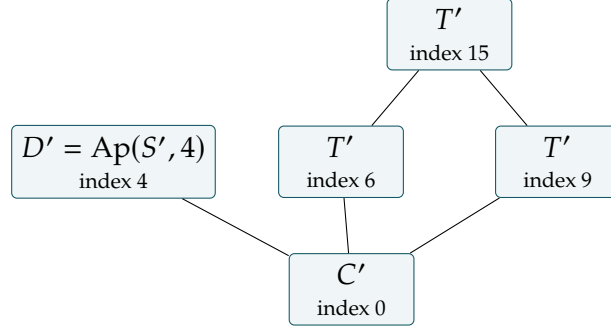


Figure 2: The exact residual block order for $\langle 4, 6, 9 \rangle$. Each edge means that every element of the lower block lies below every element of the upper block. The poisoned point is removed only from the block at index 0. Compare [figure 1](#), the same diagram with the indices 0, 4, 5, 6, 11 of $\langle 4, 5, 6 \rangle$.

Proof of theorem 1.2. Part (i). For $\sigma = 0$ the board is $\{0\}$, which has no legal Chomp move, so it is $\mathcal{P}$. For $\sigma = 1$ it is [theorem 8.2](#).

Part (ii). [Theorem 8.3](#) below proves that every positive option of T' has value at least 3. The move at 0 has value 0, and the move at 4 has value 3 by [equation \(8.5\)](#). Hence

$$\mathcal{A}(T') \cap \{0, 1, 2, 3\} = \{0, 3\},$$

which with $g(C') = 0$ is exactly [equation \(3.7\)](#). All option values of T' are finite, since every option is a finite poset. [Theorem 3.5](#) therefore applies with $a = 4$, $p = 6$, $q = 9$, $r = 15$; alternatively [theorem 3.6](#) applies, since 4 is an atom. Either way $g((T' + T') : T') = 3$ by [theorem 3.3](#), so the game in parentheses in [equation \(8.6\)](#) has value $3 \oplus_{\mathbb{N}} 3 = 0$, and $g(R') = g(C') = 0$ by [theorem 3.1](#). Thus $\omega 4 + 4$ leaves a $\mathcal{P}$ -position and wins.

Part (iii). Let $\sigma > 2$. Every $\beta \in S'^{\sigma} \setminus S'^2$ has leading exponent at least 2, so $(\omega 4 + 4) + \beta = \beta$ and therefore $\beta - (\omega 4 + 4) = \beta \in S'^{\sigma}$. The move deletes every such β and leaves exactly the residual game of the square.

For minimality, no positive *finite* opening wins: such a move deletes every ordinal with a positive higher coefficient and leaves $\text{Ap}(S', b) \setminus \{0\}$, which is $\mathcal{N}$ by [theorem 8.2](#). The only remaining legal opening below $\omega 4 + 4$ is $\omega 4$. After it the block at index 4 is absent, and the opponent answers $\omega 15$, deleting the top block and leaving $C' : (T' + T')$, which is $\mathcal{P}$ because $g(C') = 0$ and $g(T' + T') = 1 \oplus_{\mathbb{N}} 1 = 0$. There are no intermediate legal coefficients 1, 2, or 3, since those are gaps of S' . Hence $\omega 4 + 4$ is the least winning opening, and $\text{ch}(\{4, 6, 9\}) = 2$. $\square$

8.4 The poisoned–unpoisoned dictionary

Two conventions for the same boards are in circulation, and the files shipped with this report use both. It is worth fixing the translation once, because every number quoted in [section 8.7](#) from the ported certificate is a poisoned number.

Throughout this article, $g(Q)$ is the *unpoisoned* normal-play value: the board is played with all of its listed elements, including its least element 0 when present. The certificates `certificates/s456.json` and `certificates/s469.json` follow this convention, with $K = 2$ and the saturation symbol $*$ written as the digit 3, so that their labels mean

$$0 = \text{exactly } 0, \quad 1 = \text{exactly } 1, \quad 2 = \text{exactly } 2, \quad * = \text{at least } 3.$$

The certificate `certificates/s469_poisoned.json`, ported from the second source report, instead records the poisoned boards $Q \setminus \{0\}$, with the legend

$$0 = \text{exactly } 0, \quad 1 = \text{exactly } 1, \quad 2 = \text{at least } 2.$$

In particular a poisoned label 2 does *not* assert an exact value of 2. By [theorem 3.2](#) the two conventions are related by

$$g(Q) = 1 + g(Q \setminus \{0\}) \quad (8.7)$$

for every finite board Q with least element 0. Writing

$$q_x(C) = \min(2, g(P(x, C) \setminus \{0\}))$$

for the poisoned truncated entry beside the unpoisoned $v_x(C)$ of [section 5](#), the label correspondence is

$$q_x(C) = 0 \leftrightarrow v_x(C) = 1, \quad q_x(C) = 1 \leftrightarrow v_x(C) = 2, \quad q_x(C) = 2 \leftrightarrow v_x(C) = *. \quad (8.8)$$

Every poisoned number quoted below is one less than its unpoisoned counterpart, and the saturation symbols correspond.

One consequence is *not* a shift, and must be stated separately. The far spectrum [equation \(5.5\)](#) equals $\{0\}$ in the unpoisoned convention, as recorded in [equation \(6.4\)](#), because the unpoisoned move at $y = 0$ leaves the empty game, of value 0. In the poisoned convention that move does not exist, the far prefix contributes nothing below the cutoff, and the corresponding set is *empty*. The statements “ $B_x = \{0\}$ ” and “ $B_{23} = \emptyset$ ” are the same fact in the two conventions; neither contradicts the other. The same care is needed for the counts reported in [section 8.8](#).

8.5 The seventeen gap ideals, in two orders

Give $\mathcal{H}'$ the order induced by semigroup differences. Its cover relations are

$$1 < 5, \quad 1 < 7, \quad 3 < 7, \quad 2 < 11, \quad 5 < 11, \quad 7 < 11, \quad (8.9)$$

and its downsets number 17. Encode a downset $C \subseteq \mathcal{H}'$ by

$$m(C) = \mathbf{1}_{1 \in C} + 2 \cdot \mathbf{1}_{2 \in C} + 4 \cdot \mathbf{1}_{3 \in C} + 8 \cdot \mathbf{1}_{5 \in C} + 16 \cdot \mathbf{1}_{7 \in C} + 32 \cdot \mathbf{1}_{11 \in C}.$$

[Table 1](#) lists all seventeen as explicit sets, together with their masks.

Reconciling the two column orders. The two shipped certificates for this semigroup order these seventeen downsets *differently*, and this is the one place where a careless comparison produces a spurious contradiction. The file `certificates/s469.json` lists ascending masks,

$$0, 1, 2, 3, 4, 5, 6, 7, 9, 11, 13, 15, 21, 23, 29, 31, 63, \quad (8.10)$$

while `certificates/s469_poisoned.json` and [section B](#) use increasing cardinality and then lexicographic order. With j and i as in [table 1](#), the dictionary is $i = \pi(j)$ for

$$\pi = (0, 1, 2, 4, 3, 5, 7, 8, 6, 9, 10, 12, 11, 13, 14, 15, 16). \quad (8.11)$$

i	C_i	$m(C_i)$	j	i	C_i	$m(C_i)$	j
0	$\emptyset$	0	0	9	$\{1, 2, 5\}$	11	9
1	$\{1\}$	1	1	10	$\{1, 3, 5\}$	13	10
2	$\{2\}$	2	2	11	$\{1, 3, 7\}$	21	12
3	$\{3\}$	4	4	12	$\{1, 2, 3, 5\}$	15	11
4	$\{1, 2\}$	3	3	13	$\{1, 2, 3, 7\}$	23	13
5	$\{1, 3\}$	5	5	14	$\{1, 3, 5, 7\}$	29	14
6	$\{1, 5\}$	9	8	15	$\{1, 2, 3, 5, 7\}$	31	15
7	$\{2, 3\}$	6	6	16	$\mathcal{H}' = \{1, 2, 3, 5, 7, 11\}$	63	16
8	$\{1, 2, 3\}$	7	7				

Table 1: The seventeen admissible gap tails of $\langle 4, 6, 9 \rangle$. The index i sorts them by cardinality and then lexicographically, which is the column order of `certificates/s469_poisoned.json` and of [section B](#). The index j sorts them by ascending mask, which is the column order of `certificates/s469.json` and of `data/s469_low_rows.csv`.

The permutation transposes exactly three pairs of columns, namely $\{3\} \leftrightarrow \{1, 2\}$, $\{1, 5\} \leftrightarrow \{2, 3\}$, and $\{1, 3, 7\} \leftrightarrow \{1, 2, 3, 5\}$, and fixes the other eleven. Under [equation \(8.11\)](#) together with the label dictionary [equation \(8.8\)](#), the two certificates agree in every one of their $253 \cdot 17 = 4301$ entries, and their untruncated companion tables `data/s469_exact_finite_audit.csv` and `data/s469_poisoned_boundary_values.csv` agree in all 4301 exact values, the second being the first minus one throughout. The verifier code `verify_poisoned_certificate.py` asserts this agreement cell by cell on every run. A reader who compares two rows without applying [equation \(8.11\)](#) and [equation \(8.8\)](#) and finds a discrepancy has found an artefact of the two orderings, not a disagreement between the certificates.

8.6 Initial values and the specialized recurrence

There are five positive semigroup elements below the conductor. Their exact Apéry values, in both conventions, are

$$\begin{array}{c|ccccc}
 t & 4 & 6 & 8 & 9 & 10 \\
 \hline
 g(\text{Ap}(S', t) \setminus \{0\}) & 2 & 4 & 3 & 6 & 5 \\
 g(\text{Ap}(S', t)) & 3 & 5 & 4 & 7 & 6
 \end{array} \tag{8.12}$$

The second row is the first plus one, by [equation \(8.7\)](#). The file `data/s469_poisoned_apery_values.csv` carries both rows, for these five parameters and for every x up to 264; it is the only shipped artefact presenting the two conventions side by side, and it is the machine-readable form of [section 8.4](#). In particular no exceptional parameter has unpoisoned value below 3.

The eleven initial rows $12 \leq x \leq 22$ are computed directly from the finite boards $P(x, C)$, without the moving-boundary recurrence. The largest integer appearing in those initial boards is 33.

Because every exceptional value already meets the cutoff, the poisoned far-prefix set B_{23} is empty, which by [section 8.4](#) is the same statement as the unpoisoned $B_x = \{0\}$ of [equation \(6.4\)](#). The recurrence [equation \(5.6\)](#) therefore specializes, in the poisoned convention, to

$$q_x(C) = \text{mex}_2 \left(\{q_{x-d}(\Delta_d(C)) : 1 \leq d \leq 11\} \cup \{q_x(C_c) : c \in C\} \right), \quad x \geq 23, \tag{8.13}$$

where $\text{mex}_2 A = \min(2, \text{mex } A)$, and Δ_d and $C \mapsto C_c$ are the maps of [equations \(5.3\)](#) and [\(5.4\)](#). The poisoned sources call these two maps T_d and R_a ; they are the same maps.

This simplification is *checked inductively*; it is not a prior assumption of the conclusion. Every computed full-tail coordinate must remain 2 throughout the initial segment and throughout the certified cycle, and the verifier fails otherwise.

8.7 The certificate and its closure

Write the seventeen-entry poisoned row as the word

$$w_x = q_x(C_0)q_x(C_1) \cdots q_x(C_{16}) \in \{0, 1, 2\}^{17},$$

in the column order of [table 1](#); its final digit is the full-tail label. The certificate consists of the 253 words $w_{12}, \dots, w_{264}$ printed in [section B](#), and has three closure properties:

$$\begin{aligned} q_x(C_{16}) &= 2 & (12 \leq x \leq 264), \\ (w_{226}, \dots, w_{236}) &= (w_{254}, \dots, w_{264}), \\ w_x \text{ satisfies } \text{equation (8.13)} & & (23 \leq x \leq 264). \end{aligned} \tag{8.14}$$

The middle line compares exactly $11 \times 17 = 187$ labels. It says that the complete boundary state before row 237 equals the complete boundary state before row 265, that is,

$$\boxed{\mathcal{S}_{237} = \mathcal{S}_{265}}, \tag{8.15}$$

and the period is $265 - 237 = 28$.

Lemma 8.3 (Certified spectral gap for $\langle 4, 6, 9 \rangle$). *For every $t \in S' \setminus \{0\}$,*

$$g(\text{Ap}(S', t) \setminus \{0\}) \geq 2, \tag{8.16}$$

equivalently $g(\text{Ap}(S', t)) \geq 3$: every positive option of the unpoisoned game T' has Grundy value at least 3.

Proof. The finite recursion verifies [equation \(8.12\)](#), the eleven initial rows, and [equation \(8.14\)](#). Each initial full-tail label is 2, so $B_{23} = \emptyset$; inductively the specialized recurrence [equation \(8.13\)](#) is the general recurrence [equation \(5.6\)](#) and preserves the empty prefix set through row 264. The two complete boundary states then coincide, and [theorem 5.2](#) applies: determinism makes every subsequent state repeat the same cycle, in which each full-tail label is again 2. Hence $q_t(\mathcal{H}') = 2$ for every $t \geq 12$, and since $\text{Ap}(S', t) = P(t, \mathcal{H}')$ for $t > F'$ by [theorem 5.1](#), the poisoned Apéry board has value at least 2 for all such t . The five exceptional parameters below 12 are covered by [equation \(8.12\)](#). The unpoisoned reformulation is [equation \(8.7\)](#). $\square$

The row sequence is periodic from row 226 onward with period 28. The ported verifier additionally checks $w_{225} \neq w_{253}$, so 226 is the *earliest* onset for this particular period; no assertion is made about the least possible period. The analogous onset for $\langle 4, 5, 6 \rangle$, period 8 from row 36, is recorded but is not certified minimal, since `code/verify.py` performs no such check.

8.8 What was independently checked

Beyond the generator-based checker and the untruncated audit described in [section 9](#), this semigroup is examined by a third implementation, working in the poisoned convention.

That verifier rebuilds semigroup membership from the generators 4, 6, and 9 rather than from a gap list; enumerates the seventeen downsets by a deliberately different route, `itertools.combinations` rather than bit masks; and computes *untruncated* integer Grundy values with bit masks. It directly solves all $253 \times 17 = 4301$ boards represented in the certificate, not merely the eleven initial rows; its cache then holds 4346 distinct finite *poisoned* states. The largest board element used anywhere in these checks is $264 + 11 = 275$. It verifies 233 literal local residual-poset identities at $x = 23$ and cross-checks them against the certificate’s stored transition index tables; it then *replays* the bounded recurrence from those stored indices alone, consulting no exact high value and no semigroup arithmetic. It checks the 187-entry repeated state, the onset inequality $w_{225} \neq w_{253}$, the diamond, asserting that its option set is exactly $\{0, 1, 2\}$ and not merely that its value is 3, the finite gadget of [section 8.9](#), the column dictionary [equation \(8.11\)](#), and the SHA-256 digest of the certificate bytes.

Conventions attach to counts. All counts in the preceding paragraph are *poisoned*-convention counts and must be read as such. In particular the 4346 memoized states there and the 4347 distinct finite positions reported by `code/independent_check.py` in [section 9](#) are not in conflict: the first solver works on boards with 0 deleted, the second on boards with 0 retained. The same applies to the counts 233, 187, 275, and 33, and to the pair 403/4347 in the table of [section 9](#).

The regression suite. Six tests accompany the ported verifier, and they have no counterpart in the first source report. Among them, `test_general_boundary_recurrence` validates the identities behind [theorem 5.2](#) on eight numerical semigroups,

$$\langle 2, 3 \rangle, \langle 3, 4 \rangle, \langle 3, 5 \rangle, \langle 3, 4, 5 \rangle, \langle 4, 5, 6, 7 \rangle, \langle 4, 6, 9 \rangle, \langle 5, 6, 7 \rangle, \langle 6, 7, 8, 9 \rangle,$$

at the three cutoffs 1, 2, 3; that is evidence for the general theorem rather than for one semigroup. `test_capped_mex_identity` machine-checks [equation \(5.2\)](#) over all 256 subsets of $\{0, \dots, 7\}$ and cutoffs 1 through 5, an identity asserted but never machine-checked in the first source report. `test_corrupted_label_rejected` and `test_corrupted_transition_rejected` are negative controls, proving that the verifier is able to fail. `test_regeneration` pins the generator’s output against the shipped JSON. The recorded run is `data/s469_poisoned_tests.txt`.

Two generators, one of which searches. The certificate builder `code/build_certificates.py` intentionally replays the documented candidate repeat indices; it is a reproducible generator, not a search program. The ported `code/make_poisoned_certificate.py` is a search: it walks forward keeping a dictionary of complete boundary states, *discovers* the first repetition at $237 = 265$ rather than being told where it is, and then derives the onset $226 = 1 + \max\{x : w_x \neq w_{x+28}\}$. It is therefore the shipped code that underwrites the onset claim.

The trust boundary. The mathematical trust boundary consists of ordinary mathematical reasoning plus finite exact integer computations under Python. No floating-point arithmetic, random search, external solver, or conjectured period is involved. No formal proof-assistant verification is claimed. The several implementations are separate implementations of the same finite calculation, not two independent human proofs; they reduce coding risk without substituting for a proof-assistant kernel or for independent human review.

8.9 A purely finite analogue of the three-block architecture

[Theorem 3.3](#) is an algebraic statement about option spectra. It admits a purely finite confirmation on an explicit finite poset, which the ported verifier performs and which is logically independent of the semigroup calculation. This is a genuinely different confirmation of the whole three-block architecture, and it is not a truncation of an infinite board; nothing in the first source report's code does it.

Let H be the disjoint union of two two-element chains, and let $P = \mathbf{1} : H$ be a minimum placed below both chains, so that $H = P \setminus \{0\}$. Then

$$g(H) = 2 \oplus_{\mathbb{N}} 2 = 0, \quad g(P) = 1, \quad \mathcal{A}(P) = \{0, 3, 4\},$$

so P has exactly the low spectrum $\mathcal{A}(P) \cap \{0, 1, 2, 3\} = \{0, 3\}$ demanded by [equation \(3.5\)](#). Forming $K = (P + P) : P$ gives

$$g(K) = 3,$$

which is [equation \(3.8\)](#) realized on a fifteen-point poset instead of on an infinite board. Finally, writing D' for the literal four-point incidence matrix of $\text{Ap}(S', 4)$ computed from the generators,

$$g(H : (D' + K)) = 0, \tag{8.17}$$

which is [equation \(3.10\)](#) on a finite poset with the block order of [figure 2](#). The value 0 is recorded as `finite_analogue_residual_value` in `data/s469_poisoned_verification.json`.

8.10 Concrete replies after the winning move

[Section 7](#) gives a general three-phase invariant argument that names no ordinal moves, and it applies here verbatim after replacing the diamond labels 5, 6, 11 by 6, 9, 15. The catalogue below is a second and deliberately concrete derivation of the same replies: it names the actual ordinal moves, and it is retained because it exercises the block decomposition [equation \(8.6\)](#) directly, whereas the invariant argument establishes only that a suitable finite move exists. Both are kept; neither supersedes the other.

Let Alice have opened at (4, 4). Bob now moves in one of the five blocks of [figure 2](#).

Bob moves in the bottom block, $a = 0$. Every upper block disappears. The residual is a finite poisoned Apéry board, of value at least 2 by [theorem 8.3](#). Alice computes an ordinary finite-poset move to value 0 and thereafter restores value 0 after each reply.

Bob moves in the diamond block, $a = 4$. Let $d \in \{0, 1, 2\}$ be the remaining unpoisoned Grundy value of that block. Alice changes the three-block component $K' = (T' + T') : T'$ to the same value d :

d	Alice's move	New three-block value
0	(15, 0)	$g(T' + T') = 0$
1	(6, 0)	$g(T') = 1$
2	(6, 4)	$g(D' + T') = 3 \oplus_{\mathbb{N}} 1 = 2$

The two independent upper components then cancel.

Bob moves in a middle block, $a = 6$ or $a = 9$. The top block at index 15 disappears. If Bob chose inner 0, Alice plays $(4, 15)$, leaving the diamond branch at value 1, which cancels the untouched middle block. Otherwise Bob has left a finite unpoisoned Apéry board of value $b \geq 3$; Alice moves inside that same finite block to leave value 2, which is possible because $2 < b$. The three remaining independent upper values are then 2, 1, and 3, whose exclusive-or is 0.

Bob moves in the top block, $a = 15$. If he chose inner 0, Alice removes the whole diamond by playing $(4, 0)$, and the two untouched middle blocks cancel. Otherwise the finite top residual has unpoisoned value at least 3, and Alice moves there to leave value 1; the excluded-ordinal formula [theorem 3.1](#) restores the three-block value to 3, which again cancels the diamond.

Every described reply leaves a position of value 0, and the finite moves to a requested smaller number are obtained by the same exact finite-poset recursion used in the verifiers. This response analysis is not a claim of a uniform polynomial-time strategy for arbitrary later ordinal-game positions; the existence of the winning strategy already follows from the proved zero-value residual together with termination.

8.11 What is established, and what is not

The result settles the stated minimal-transition question negatively and refutes winner stability: a numerical-semigroup generator set can have transition exponent strictly larger than its first admissible exponent. It does not contradict the positive result for losing semigroups of maximal embedding dimension in [1, Proposition 4.2]. Both examples here have multiplicity 4 and three minimal generators, rather than four minimal generators.

The universal assertion actually established is a low-value exclusion,

$$g(\text{Ap}(S, t) \setminus \{0\}) \notin \{0, 1\} \quad (t > 0),$$

for each of the two semigroups. The raw Grundy values are not claimed to be periodic. In particular the label 2 in the last column of the ported certificate does *not* mean that every Apéry board has value 2: the early exact values 2, 4, 3, 6, 5 of [equation \(8.12\)](#) already show otherwise. Nor is the exponent 2 obtained by testing finitely many exponents. It is established by proving a losing position at exponent 1, giving an explicit winning move at exponent 2, and proving that this same move removes every additional higher-order element.

We do not claim that these are the smallest counterexamples under every possible measure of semigroup size. The minimality proved here concerns the *winning ordinal move* and the exact transition for these two specified semigroups. We have not classified all semigroups with transition 2, established a larger transition, or proved that an unrestricted ordinal-level winner problem is decidable. A negative answer to the minimal-transition question does not itself imply undecidability.

8.12 Questions left open

A natural next classification problem is to determine which numerical semigroups have first winning exponent 2 and which preserve a second-player win at every ordinal exponent. [Theorem 3.6](#) supplies a searchable sufficient condition, but it is not a characterization.

A second question is whether the transition exponents of finite natural-number generating sets are bounded by a small finite ordinal. Nothing here establishes such a bound or constructs an exponent above 2.

On the finite-state side, one can ask for a conceptual, non-computational proof of the specific excluded value that would replace the certificate, or for invariants that *compress* its 187-entry state. The present proof needs no such compression: all entries and the transition law are already finite and explicit. The finite-window theorem [theorem 5.2](#) also suggests systematic searches for other small option-spectrum patterns that trigger an ordinal winner change, and it identifies a reusable approach, since membership of any fixed finite nimber in the one-move spectrum is effectively decidable. These are further questions, not assumptions in the present proof.

9 Reproducibility and verification boundaries

The archive contains the article source and PDF, three JSON certificates, CSV tables, a minimal verifier, a general verifier, an independent finite audit, a poisoned-convention verifier with its generator, table exporter and regression suite, and the recorded verification output. No external Python package is required, and no network access is used. The scripts are written for Python 3.9 or later, with the unpoisoned scripts assuming 3.10 or later. The runs recorded in `verification.txt`, `verification_poisoned.txt` and `data/s469_poisoned_tests.txt` used Python 3.14.4 on this merged package; the two source reports recorded their original runs under Python 3.13.5.

9.1 Commands

From the archive root, the unpoisoned checks are

```
python code/minimal_verifier.py
python code/verify.py
python code/independent_check.py
```

The first program is the complete small recurrence check printed in [section C](#). The second recomputes the gap sets from the given generators, constructs every admissible gap downset, verifies the seed rows and each recurrence row, checks the exceptional parameters, and compares the complete repeated states. The third does not import the recurrence implementation: it builds principal upper sets from the generators, encodes finite positions as bit masks, and computes *full* Grundy numbers by literal move enumeration. Both of the last two scan the certificates directory for names matching the pattern `s4dd.json` only, so that the poisoned certificate described below is not handed to a verifier expecting the unpoisoned schema.

The poisoned-convention checks, which concern $\langle 4, 6, 9 \rangle$ only, are

```
python code/verify_poisoned_certificate.py
python code/test_poisoned_certificate.py
```

and the two regeneration steps are

```
python code/build_certificates.py
python code/make_poisoned_certificate.py
python code/export_poisoned_tables.py
```

The targets `make verify`, `make test`, `make certificates` and `make pdf` wrap these. `build.ps1`

runs the verifiers, the test suite and the three $\text{\LaTeX}$ passes on Windows PowerShell, but not the regeneration step, so `make certificates` has no PowerShell counterpart. That PowerShell wrapper is supplied for convenience; it was not executed in the Linux verification environment in which the original poisoned checks were run.

9.2 The three certificates

Three certificate files are shipped, for two semigroups.

File	Semigroup	Convention	Column order
<code>certificates/s456.json</code>	$\langle 4, 5, 6 \rangle$	unpoisoned, $\ast = 3$	ascending masks
<code>certificates/s469.json</code>	$\langle 4, 6, 9 \rangle$	unpoisoned, $\ast = 3$	ascending masks
<code>certificates/s469_poisoned.json</code>	$\langle 4, 6, 9 \rangle$	poisoned, $2 = \text{“at least 2”}$	cardinality, then lex

The last two describe the *same* 253 rows over the same seventeen gap ideals; [section 8.5](#) gives the exact dictionary, and `code/verify_poisoned_certificate.py` checks it. Both are kept: the poisoned file is the only one carrying the transition index tables `transition_recent_indices` and `transition_tail_indices`, the fields `label_meaning`, `convention`, `conductor`, `frobenius`, `cutoff`, `period_start` and `early_aper_y_values`, and the published digest.

9.3 The digest

The poisoned certificate has

```
SHA-256 = f408697a728c12efd4cca15b73d055b3c747a17e5b06a5897969973ceb1697dc,
```

recorded in `data/s469_poisoned_verification.json`, `verification_poisoned.txt` and the README. This digest stays valid only while the file is preserved byte for byte; any reformatting, re-indentation, key reordering or change of line endings would silently invalidate it. The generator therefore writes the file with explicit `"\n"` line endings, so that regeneration on any platform reproduces the same bytes, and the verifier recomputes and prints the digest on every run. Repository-wide `SHA256SUMS.txt` manifests are no longer distributed, so this is the only per-file integrity claim in the archive. Recompiling the document may change the PDF’s byte hash, because of timestamps and TeX versions; the JSON certificates are deterministic.

9.4 What the executed checks reported

Semigroup	Convention	Certificate entries	Direct finite states	State repetition
$\langle 4, 5, 6 \rangle$	unpoisoned	387	403	43 = 51
$\langle 4, 6, 9 \rangle$	unpoisoned	4301	4347	237 = 265
$\langle 4, 6, 9 \rangle$	poisoned	4301	4346	237 = 265

“Direct finite states” counts distinct positions visited by the corresponding independent recursive evaluator, not all possible infinite-game states. The two counts for $\langle 4, 6, 9 \rangle$ differ by exactly the convention, as explained in [section 8.8](#): 4347 unpoisoned boards against 4346 poisoned ones.

The files `verification.txt` and `verification_poisoned.txt` record the actual outputs, and `data/s469_poisoned_tests.txt` records the regression run, with all six tests passing.

9.5 Why a finite certificate proves an infinite assertion

There are three separate verification obligations. The code checks the finite values and the repeated states. The proof of [theorem 5.2](#) explains why a repeated state covers all larger parameters. Finally, the order decomposition and the game algebra, or equivalently the response procedures of [sections 7](#) and [8.10](#), connect the spectral fact to the ordinal opening. A direct finite audit alone would not replace either of the latter two mathematical arguments.

It is worth stating the negative version explicitly. No assertion about an infinite board is inferred merely from a large finite truncation. The complete state is an eleven-row window for $\langle 4, 6, 9 \rangle$ and a seven-row window for $\langle 4, 5, 6 \rangle$, together with the accumulated far spectrum; once that window repeats, the same deterministic operation repeats forever. The exact, unbounded Grundy sequence is not claimed to be eventually periodic; only each fixed bounded profile is, as [theorem 5.3](#) records.

To rebuild the document, use `make pdf`, which runs `pdfLATEX` three times to settle the table of contents and cross-references, or run `build.ps1`. The numerical certificates are independent of the document-build process.

A The complete certificate for $\langle 4, 5, 6 \rangle$

Column labels are the shape masks in [equation \(6.1\)](#); * denotes a Grundy value at least 3, so these are *unpoisoned* labels. The first seven rows are seeds. The shaded rows 36–42 repeat at 44–50; the row at 43 is computed from the first repeated window. Every last-column entry is saturated. The same data are in `certificates/s456.json`.

x	0	1	2	3	4	5	6	7	15
8	2	*	*	2	*	2	2	*	*
9	*	2	1	*	*	*	2	1	*
10	2	*	*	*	*	*	2	1	*
11	*	*	*	*	*	*	*	*	*
12	2	1	*	2	*	2	2	1	*
13	*	2	2	*	*	*	*	*	*
14	2	*	*	*	*	*	2	1	*
15	*	*	*	1	*	*	*	*	*
16	2	*	*	2	*	2	2	*	*
17	*	2	2	*	*	*	*	*	*
18	2	*	*	*	*	*	2	*	*
19	*	*	*	*	*	*	*	*	*
20	2	1	*	2	*	2	2	1	*
21	*	2	2	*	*	*	*	*	*
22	2	1	*	*	*	*	2	1	*
23	1	*	*	1	*	1	1	*	*
24	2	*	*	2	*	1	2	*	*
25	*	2	1	*	*	*	2	*	*
26	2	*	*	*	*	2	2	*	*
27	*	*	*	*	*	*	*	*	*
28	2	1	*	*	*	*	2	1	*
29	*	2	2	*	*	*	*	*	*
30	2	1	*	2	*	*	2	1	*
31	1	2	*	1	*	1	1	*	*
32	2	*	*	2	*	1	2	*	*
33	*	*	1	*	*	*	*	*	*
34	2	*	*	2	*	2	2	*	*
35	*	*	*	*	*	*	*	*	*
36	2	1	*	*	*	*	2	1	*
37	*	*	*	*	*	*	*	*	*
38	2	1	*	2	*	2	2	1	*
39	1	2	2	1	*	1	1	*	*
40	2	*	*	*	*	1	2	*	*
41	*	*	1	*	*	*	*	*	*
42	2	*	*	2	*	2	2	*	*
43	*	2	2	*	*	*	*	*	*
44	2	1	*	*	*	*	2	1	*
45	*	*	*	*	*	*	*	*	*
46	2	1	*	2	*	2	2	1	*
47	1	2	2	1	*	1	1	*	*
48	2	*	*	*	*	1	2	*	*
49	*	*	1	*	*	*	*	*	*
50	2	*	*	2	*	2	2	*	*

B The complete certificate for $\langle 4, 6, 9 \rangle$

Each word has seventeen digits, in the column order i of [table 1](#), which is *not* the mask order used in [section A](#); see [equation \(8.11\)](#). These are *poisoned* labels, so a digit 2 means “Grundy value at least 2” for the board with 0 deleted, equivalently at least 3 for the unpoisoned board; see [equation \(8.8\)](#). The final digit of each word is the full-tail label. The first eleven words, $w_{12}, \dots, w_{22}$, are the initial conditions; every subsequent word follows from [equation \(8.13\)](#). The blocks 226–236 and 254–264 are identical, which is [equation \(8.15\)](#). The table is printed in three independent index–word columns, read downward within each column. The same data are in `certificates/s469_poisoned.json`, and the unpoisoned form of the same rows, in mask order, is in `certificates/s469.json`.

Table 2: All 253 certified bounded-Grundy row vectors.

x	w_x	x	w_x	x	w_x
12	12222122222211102	97	10222122220222122	182	21212002212220222
13	21022222021222202	98	22222001221120222	183	22221222222222212
14	22222112222221202	99	22222222222222222	184	02222021212200022
15	22222222222222222	100	02222022222200022	185	10222222220222122
16	12222122222211122	101	20222222220222222	186	22222001222220222
17	21222222212222222	102	12221001221120222	187	22222222222222222
18	22222112222221202	103	21122222212222222	188	02222022122200012
19	22222222222222222	104	02112022222200012	189	20222222220222222
20	12222122220211102	105	20221222220212222	190	22222002222220222
21	21222222221222222	106	12221001221120222	191	22222222222222222
22	22222112222221202	107	21122222221222222	192	02221021222200022
23	22222222222222222	108	02122022222200012	193	20222222220222222
24	12222122222211122	109	20222221220222222	194	22222002222220222
25	21222222212222222	110	12222001221220222	195	22122222122222212
26	22202112222221202	111	21222222221222222	196	02212022122200012
27	22220222222202222	112	02222022222200022	197	20221222220212222
28	12222122220211102	113	20221221220222222	198	21222002111220222
29	21222222212222222	114	22222002221220222	199	22222222222222222
30	22222112222221202	115	22222222222222222	200	02221021222200022
31	22222222222222222	116	02222021222200022	201	20222222220222222
32	12222102222211102	117	20122222220222222	202	22212002212220222
33	21222222212222222	118	22221002222210222	203	22222222122222212
34	22222112222221222	119	22222222222222222	204	02222022222200022
35	22022222022222202	120	01112022122200012	205	20222222220222222
36	12222122222211102	121	20221222220212222	206	22222002222220222
37	21222222221222222	122	22212002122220212	207	22222222222222222
38	22222112220221202	123	12221122222212122	208	02122022122200012
39	22222222222222222	124	02221022222200012	209	10222221210222222
40	12222122222211102	125	20222222220222222	210	21122002221220222
41	21222222212222222	126	22222001212220222	211	22212222122222212
42	22222102212221202	127	22222222222222222	212	02222022222200022
43	22222222222222222	128	02222022222200022	213	20221222220212222
44	12222122222211102	129	20222222220222222	214	22222002222220222
45	21220222221202222	130	22222002222220222	215	21212222212221222
46	22222112222221222	131	22222222222222222	216	02222022122200012
47	22222222222222222	132	02222022121100012	217	20222221220222222
48	12222122222211102	133	20222222220222222	218	21222002221220222
49	21222222221222222	134	21222002221220212	219	22222222222222222

Continued on the next page

x	w_x	x	w_x	x	w_x
50	22222112220221202	135	2222222222222222	220	022220222222200022
51	2222222222222222	136	01212022212200022	221	20221222220212222
52	12222122222211122	137	20221222220212222	222	22222002222220222
53	21222222221222222	138	22212002212220222	223	12221121222222122
54	22222102212221202	139	12222122222222122	224	02222021222200022
55	22222222222222222	140	02222021222200022	225	20122222220222222
56	12222122220211102	141	20222222220222222	226	22221002222210222
57	21022222021222202	142	22222002121120212	227	22222222222222222
58	22222112222221202	143	22222222222222222	228	02222022222200022
59	22222222222222222	144	02222022222200022	229	20122222120222212
60	12222102222211102	145	20222222220222222	230	12222001222220222
61	21222222221222222	146	21212002212220222	231	21122222221222222
62	22222112222221202	147	22221222222212222	232	02221022222200012
63	22022222222202222	148	02212012222200012	233	20222222220222222
64	12222122222211102	149	10222122220222122	234	22122002122220212
65	21222222221222222	150	22222002122220212	235	22122222222212222
66	22222112222221202	151	22122222122222222	236	02112022222200012
67	22222222222222222	152	02221022222200012	237	20212222120222212
68	12222122222211122	153	20222222220222222	238	22222002212220222
69	21222222221222222	154	22222002222220222	239	22222222122222212
70	22222112222221222	155	22221221222222222	240	02222022222200022
71	22222222222222222	156	02222021222200022	241	20222222220222222
72	12222122220211102	157	20122222220222222	242	22222002222220222
73	21222222221222222	158	22222002222210222	243	22222222222222222
74	22222112220221202	159	22222222222222222	244	02222022122200012
75	22222222222222222	160	02222022121100012	245	20222222220222222
76	10222122202211022	161	20222222220222222	246	22221002222210222
77	21222222221222222	162	22222002221220212	247	22222222222222222
78	22202102212221122	163	22222222222222222	248	02222022222200022
79	02222022222220222	164	01222022212200022	249	20221222220212222
80	12222122220221202	165	20222222220222222	250	22222002222220222
81	21022222021222222	166	22212002212220222	251	12221121222222122
82	22222112222201222	167	12222122222222122	252	02222022222200022
83	22222222222222222	168	02222021221200022	253	20222222220222222
84	10222122202210022	169	20222222220222222	254	22221002222210222
85	21222222221222222	170	22222002221220212	255	22222222222222222
86	12222120212221122	171	22222222222222222	256	02222022222200022
87	01222022221222022	172	01222022211200022	257	20122222120222212
88	12222021222100022	173	20222222220222222	258	12222001222220222
89	21222222221222222	174	21212002212220222	259	21122222221222222
90	21222002121220212	175	12222122222222122	260	02221022222200012
91	22222222222222222	176	02222011222200022	261	20222222220222222
92	01212022212200022	177	10222122220222222	262	22122002122220212
93	10221122220212122	178	22222002121120212	263	22122222222212222
94	21222001212220122	179	22122222122222212	264	02112022222200012
95	22222222222222222	180	02221022222200012		
96	02222021212200022	181	20222222220222222		

C A self-contained verifier for the main spectral lemma

This program computes the seeds directly, evolves the seven-row state, and checks its repetition. Here 3 is a saturation marker. Run it without Python's -O, which disables assertions. The longer verifier uses explicit checks and derives the gap set from the generators.

```
# Self-contained proof certificate for S=<4,5,6>. Python 3.10+.
# Value 3 is the symbol "at least 3", not an exact number.
from functools import lru_cache

gaps = (1, 2, 3, 7)
shapes = tuple(range(8)) + (15,)

def member(n):
    return n >= 0 and n not in gaps

def elements(c):
    return {g for i, g in enumerate(gaps) if c >> i & 1}

def encode(values):
    return sum(1 << i for i, g in enumerate(gaps) if g in values)

def low_mex(options):
    return next((i for i in range(3) if i not in options), 3)

@lru_cache(None)
def value(position):
    return low_mex({value(tuple(z for z in position
                                if not member(z-y)))
                    for y in position})

def position(x, c):
    return (tuple(n for n in range(x) if member(n))
            + tuple(x+g for g in sorted(elements(c))))

def delta(d, c):
    return encode({g for g in gaps if g < d or g-d in elements(c)})

def cut(a, c):
    return encode({g for g in elements(c) if not member(g-a)})

rows = {x: {c: value(position(x, c)) for c in shapes}
        for x in range(8, 15)}
far = {}
for b in (4, 5, 6):
    apery = tuple(n for n in range(b+8)
                  if member(n) and not member(n-b))
    assert value(apery) == 3
states = {}
for x in range(15, 52):
    states[x] = (tuple(sorted(far)),
                tuple(tuple(rows[t][c] for c in shapes)
                      for t in range(x-7, x)))

    if x == 51:
        break
    row = {}
    for c in sorted(shapes, key=int.bit_count):
        near = {rows[x-d][delta(d, c)] for d in range(1, 8)}
        tail = {row[cut(a, c)] for a in elements(c)}
        row[c] = low_mex(far | near | tail)
    rows[x] = row
    if rows[x-7][15] < 3:
        far.add(rows[x-7][15])
assert all(rows[x][15] == 3 for x in rows)
assert states[43] == states[51]
print("PASS: 387 entries; state 43 = state 51; period 8.")
```

D A compact statement of the checking algorithm

Section C prints a complete program for the smaller semigroup. For $\langle 4, 6, 9 \rangle$ the analogous listing would be longer, so we state only the mathematical core of the recurrence check. Here tails is the list of gap ideals in the order of table 1, index maps each ideal to its column, rows contains the already checked words, and `member(n)` tests $n \geq 0$ and $n \notin \{1, 2, 3, 5, 7, 11\}$. The values are poisoned labels.

```
def next_row(x, rows, tails, index):
    result = []
    for C in tails:                # increasing cardinality
        options = set()
        for d in range(1, 12):
            T = frozenset(g for g in gaps
                           if g < d or g - d in C)
            options.add(rows[x - d][index[T]])
        for a in C:
            R = frozenset(g for g in C if not member(g - a))
            options.add(result[index[R]])
        value = 0
        while value in options:
            value += 1
        result.append(min(2, value))
    return tuple(result)
```

The actual verifier reconstructs the membership test from the generators 4, 6 and 9 rather than from the printed gap list. The omitted far-prefix contribution is justified by checking that the initial exceptional Apéry values and all full-tail coordinates are at least 2, as in section 8.6. The complete state is then an eleven-row window; once that window repeats, the same deterministic operation repeats forever.

For the finite seed calculation, and for the independent exhaustive check, the only game-theoretic recursion needed is the literal set-deletion form of equation (2.1):

$$g(A) = \text{mex}\{g(\{z \in A : z - y \notin S\}) : y \in A\}, \quad g(\emptyset) = 0. \quad (\text{D.1})$$

All sets A here are finite sets of nonnegative integers. Each recursive argument is a proper subset of A , so the calculation terminates by induction on $|A|$. The bit-mask verifiers implement exactly this recurrence, with exact, unrestricted nonnegative-integer values.

References

- [1] Fabián Rivero Herrera, *Hanf numbers for poset games*, arXiv:2504.07317v3, January 5, 2026. Target: Question 4.6, credited in that revision to Ignacio García-Marco; generated ordinal posets in Section 2; see also Example 3.2 for winner changes among arbitrary finite pointed posets, and Proposition 4.2 for maximal embedding dimension. <https://arxiv.org/abs/2504.07317v3>.
- [2] Fabián Rivero Herrera, *A poset game in submonoids of additively indecomposable ordinals*, arXiv:2504.07317v1, April 9, 2025. Earlier formulation: Conjecture 4.1, the winner-stability assertion refuted here. <https://arxiv.org/abs/2504.07317v1>.
- [3] Ignacio García-Marco and Kolja Knauer, *Chomp on numerical semigroups*, Algebraic Combinatorics **1** (2018), no. 3, 371–394. doi:10.5802/alco.16. Theorem 2.4 supplies the symmetric-semigroup outcome argument; Section 6 develops finite-gap locality, which theorem 5.1 refines. The study of this game originates here. Preprint: <https://arxiv.org/abs/1705.11034>. <https://alco.centre-mersenne.org/articles/10.5802/alco.16/>.
- [4] Mišo Gavrilović and Alexander Thumm, *The Ordered Join of Impartial Games*, arXiv:2104.13131v2, May 18, 2021. Section 2 treats Grundy sets and the ordinal-sum formula; see especially Theorem 2.6, on which theorem 3.2 leans, and Section 4 for the ordered-join substitution theory of theorem 3.4. <https://arxiv.org/abs/2104.13131v2>.